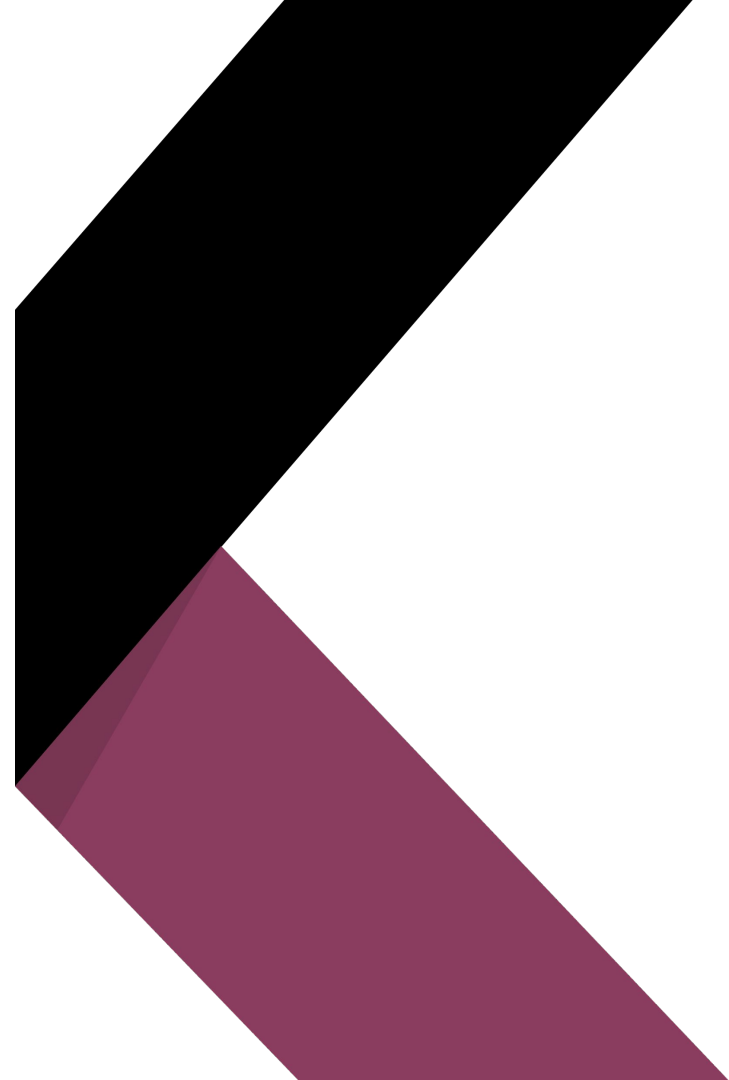


Hardware-in-the-loop

# Introduction



# Overview

- Introduction
- Why HIL?
- System architecture
- dSPACE SCALEXIO
- dSPACE ControlDesk



# Introduction

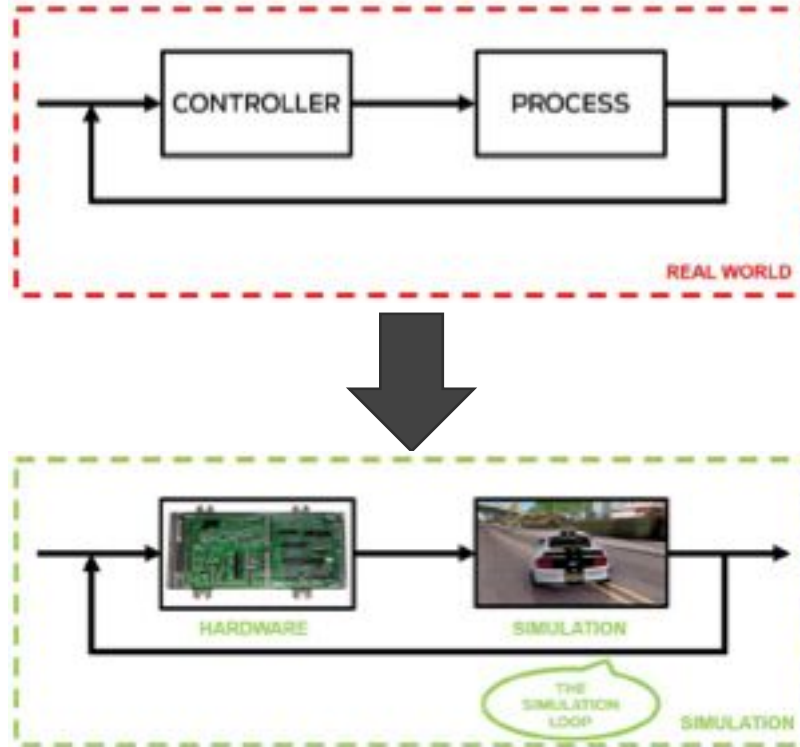


## Hardware-in-the-loop

- This history of HIL simulation in the automotive industry has its roots in the 1980s with the advent of microprocessor-based ECUs for engine controls.
- Over time HIL simulation branched out into other areas such as aerospace, military, naval, and robotics, and especially automotive industries.
- HIL simulation allows to keep the features of both **computer simulation** and **vehicle testing** by having the real controller present, while simultaneously allowing for test scenarios to be adjusted easily. In other words, HIL simulation provides a test platform in which the system or process under test consists of **actual hardware components** while the remainder of the system or process is simulated using **physics-based or mathematical models**.



## Hardware-in-the-loop



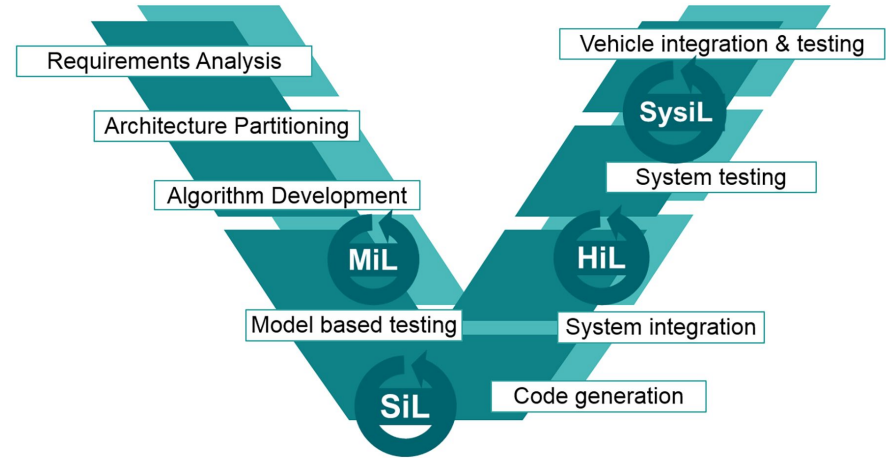


# Why HIL?



## V-Cycle

- Shorten the time to market and reduce costs of testing
- Reproducibility
- On-vehicle tests dangerous and not always in the same conditions
- Scalable system
- Critical: develop an accurate model of the real plant

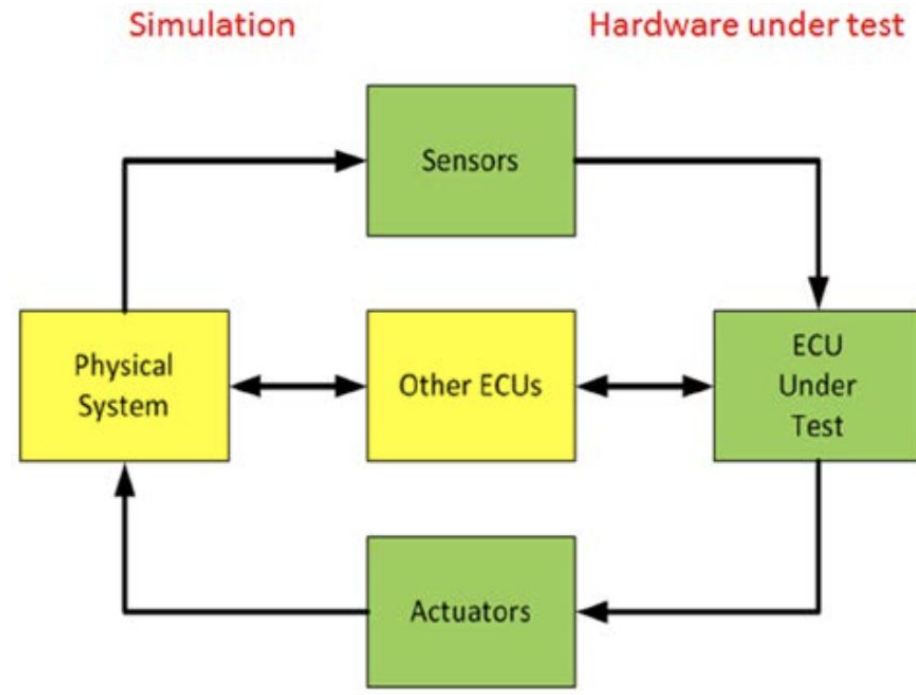




# System architecture

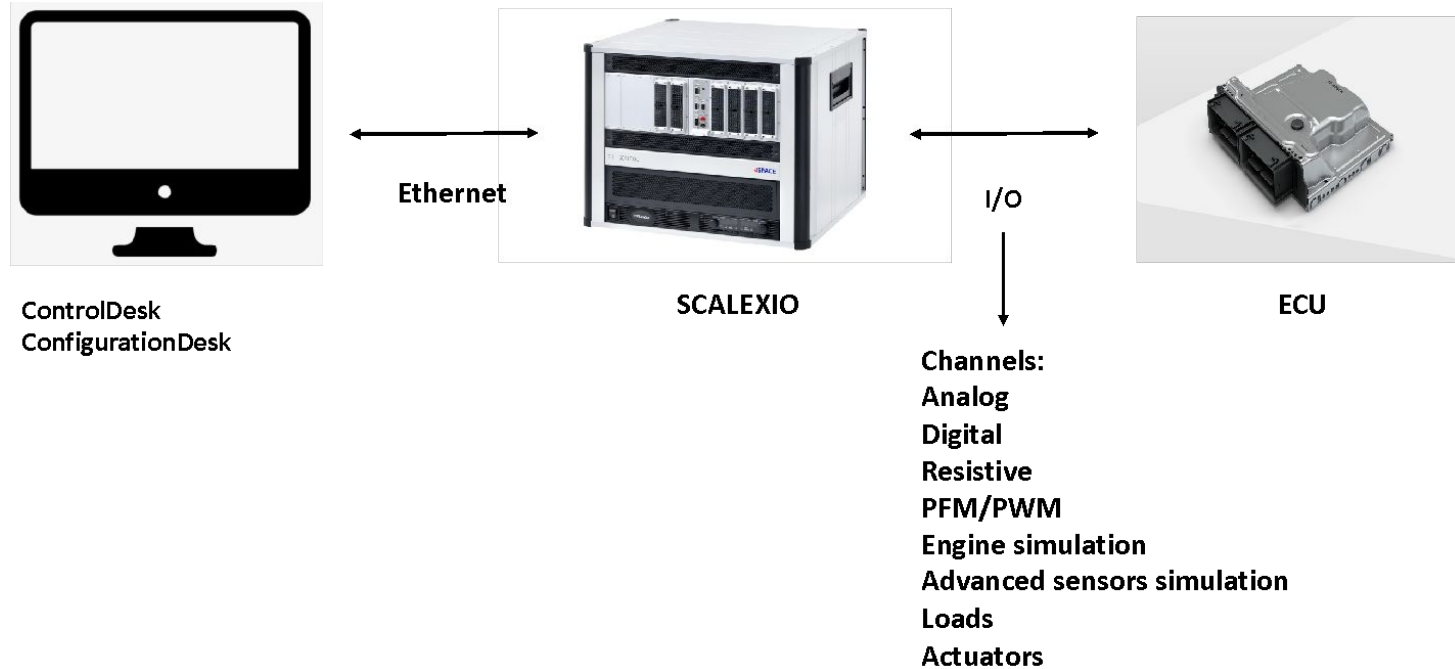


## HIL architecture





## System architecture

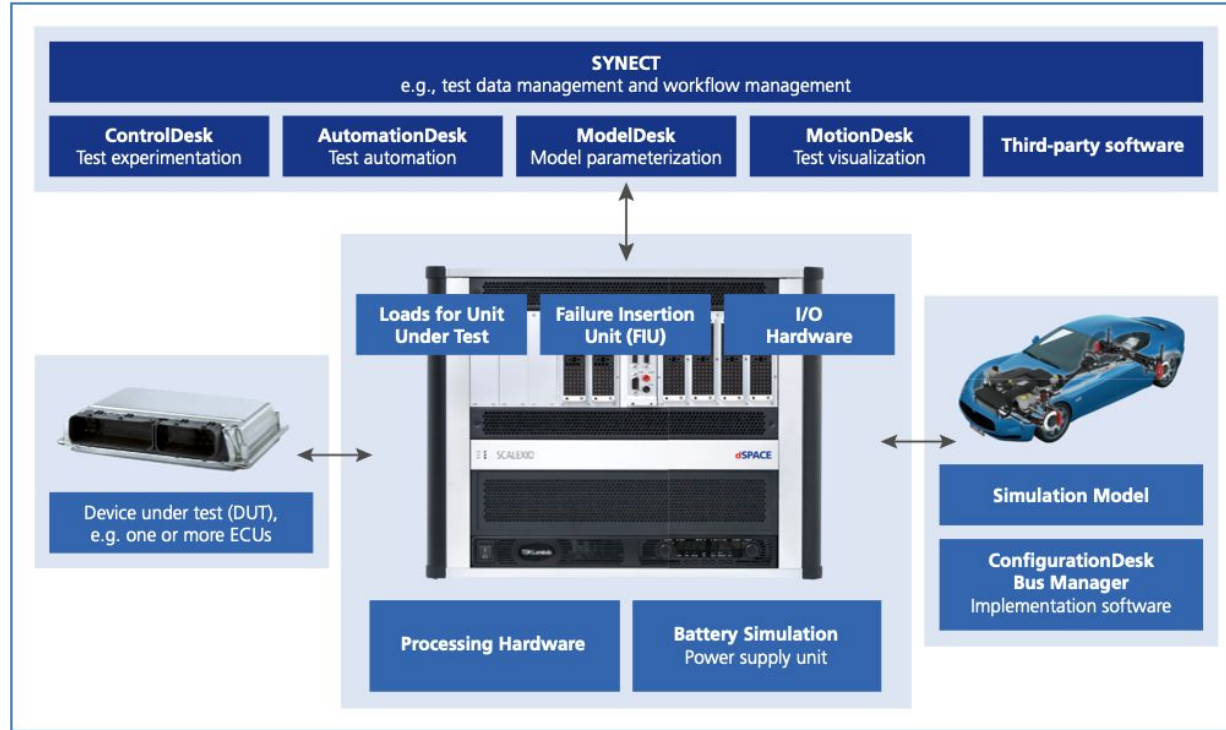




# SCALEXIO



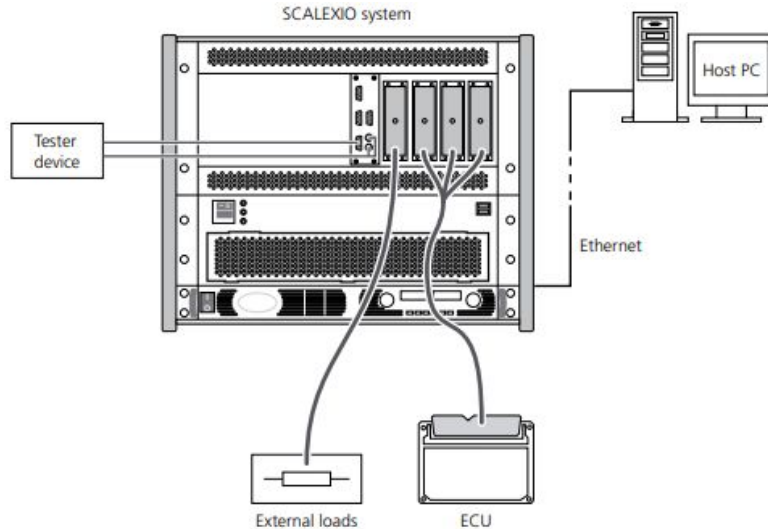
# SCALEXIO



Schematic overview of a typical HIL system and its components.



# SCALEXIO



## Features:

- Run the real-time application
- Generate sensor signals to the ECU
- Measure command signals for actuators
- Provide battery voltage to the ECU
- Simulate faults (FIU)

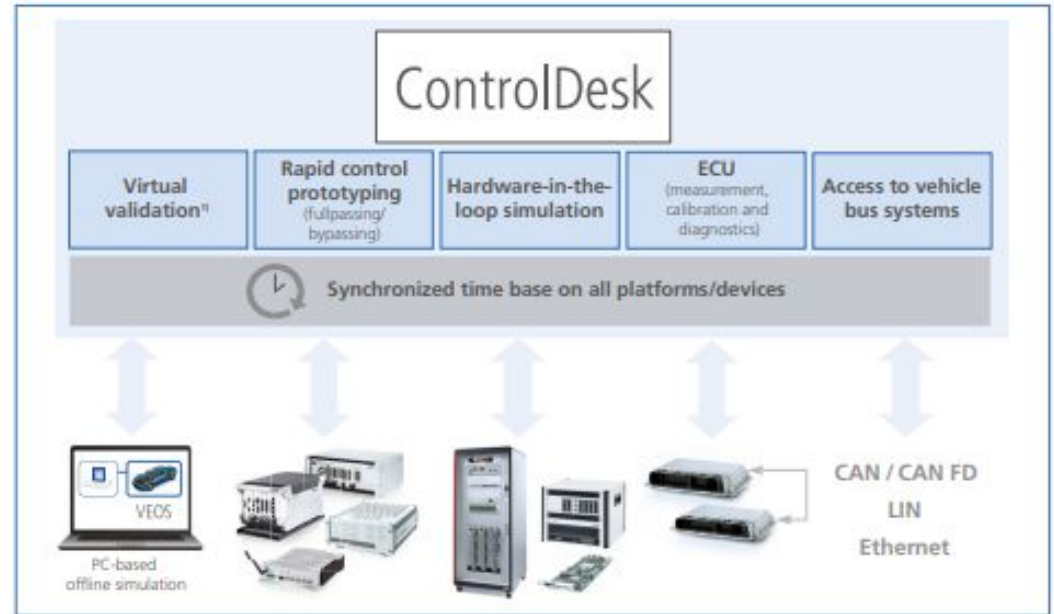


# ControlDesk



# ControlDesk

- Experimenting software for online testing
- Access to the calibration, measuring and diagnostics of the ECU
- Data acquisition synchronized on ECUs, RCP and HIL platforms and bus systems
- Enhanced layout, instrumentation, measuring and post-processing
- Support for all the dSPACE hardware and virtual validation platforms





# ControlDesk

Quick Access toolbar    Ribbon    Working area    - Instrument Selector  
- Properties

- Project Manager  
- Measurement Configuration  
- Layout Navigator  
- Instrument Navigator  
- Bus Navigator

- Variable Browser  
- Measurement Data Pool  
- Platform / Device Manager  
- Interpreter  
- Log Viewer

Status bar

- User Functions Output  
- Signal Selector  
- Signal Mapping  
- Failure Simulation  
- EESPort Configurations



# ControlDesk – Bus Navigator

| Name              | Identifier | Channel | Bustype |
|-------------------|------------|---------|---------|
| Bus Navigator     |            |         |         |
| System [CAN]      |            |         |         |
| System [Platform] |            |         |         |
| Platform [Master] |            |         |         |
| RT-CANHW          |            |         | CAN     |
| C1_Master         |            |         | CAN     |
| C2_Master         |            |         | CAN     |
| CANTX2 [body]     |            |         | CAN     |
| AIR_CONDITION     |            |         | CAN     |
| AIR_CONDITION_1   | 0x231      |         | CAN     |
| AIR_CONDITION_2   | 0x20B      |         | CAN     |
| DIAGNOSTICS       | 0xEF34576  |         | CAN     |
| IGNITION_1        | 0x001      |         | CAN     |
| CENTRAL_GATEWAY   |            |         | CAN     |
| DISPLAY           |            |         | CAN     |
| PARK_SYSTEM       |            |         | CAN     |
| TIRE_PRESSURE     |            |         | CAN     |
| C3_Master         |            |         | CAN     |
| C4_Master         |            |         | CAN     |
| ControllerGw      |            |         | CAN     |
| MainBlockGw       |            |         | CAN     |
| RT-FLEXRAYHW      |            |         | FlexRay |
| CTR0_CTR1         |            |         | FlexRay |
| CTR0_CTR1         |            |         | FlexRay |
| RT-LINHW          |            |         | LIN     |
| Controller1       |            |         | LIN     |
| Controller2       |            |         | LIN     |

No filter is active Network Node Hierarchy

The Bus Navigator displays bus configurations.

A bus configuration is the configuration of all the controllers, communication matrices, and messages/frames/PDUs of a specific communication bus such as CAN.

Simulation applications built with one of the following contain a bus configuration:

- RTI CAN MultiMessage Blockset
- RTI LIN MultiMessage Blockset

Through the Bus Navigator communication controllers, networks, nodes, messages and signals can be managed.

# CONTACTS

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